

Manoel Da Ponte

AI ENGINEER • DATA ENGINEER

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AI and data engineer, co-founder of WiseTwin: LLM agents, RAG pipelines and a multitenant SaaS platform in production. French citizen (EU/EFTA national), based in Switzerland.

[For more details and project examples, visit manoel-daponte.com](https://manoel-daponte.com)

Skills

Generative AI & Agents

LLM agents (tool use), RAG, pgvector, LLM-as-judge, SSE, multimodal vision

Data Engineering

ETL, PostgreSQL, pgvector, SQLAlchemy, Prisma, pandas, Prefect

Software Development

Python, FastAPI, TypeScript, Next.js, React, Node.js

Cloud & DevOps

Azure, Docker, GitHub Actions, Vercel, AWS

Real-Time 3D

PlayCanvas, Gaussian Splatting, GLSL shaders, Unity

Languages

French (native), English (C1)

Work Experience

WiseTwin

Dunkirk, France (remote)

CO-FOUNDER & LEAD DEVELOPER

March 2024 - Present

- Multitenant SaaS platform for industrial safety training: 80 000 lines of TypeScript, 124 API routes, SSO, MFA, xAPI/Qualiopi exports
- HSE AI copilot: LLM agent with 12 tools, pgvector RAG over 62 726 industrial accidents, SSE streaming, multimodal vision
- Automated course generation from PowerPoint: Whisper transcription, LLM writing, quizzes verified by LLM-as-judge
- Web-based 3D editors (Gaussian Splatting, Google Maps 3D); Azure Container Apps infrastructure, GitHub Actions CI/CD
- Technologies: Python, FastAPI, TypeScript, Next.js, Claude, pgvector, PostgreSQL, Docker, Azure

TotalEnergies

Esbjerg, Denmark

DATA ENGINEER

March 2022 - February 2024

- End-to-end data projects in small teams (1 to 3 people): data management, backend and frontend
- Accident prevention (IoT, NLP, LLM fine-tuning), invoice control through computer vision, large-scale SAP extraction
- Offshore IoT architecture: data warehouse, anomaly detection, Prefect monitoring (Python, Azure, SQL)

CGI

Toulouse, France

DATA SCIENTIST

July 2020 - March 2022

- Documentation chatbot with semantic search (Elasticsearch) and data migration on AWS (Python, NLP, SQL)
- Behavioral analytics and personalized recommendations to improve user journeys

Education

MSc in Data Science and Engineering

Paul Sabatier University, Toulouse III

GRADUATE

2019 - 2021

BSc in Applied Mathematics (MIASHS)

Jean Jaurès University, Toulouse II

GRADUATE

2018 - 2019

Two-year degree (DUT) in Statistics and Business Intelligence

University of Caen

GRADUATE

2016 - 2018